

Bluetooth Audio Adaptor — Design Overview

Rev B (in design) · Updated 2026-09-16 · Jason

Describes the **target design**. Deltas from as-built Rev A are listed in §10.

Living document. Every confirmed design change lands here. Planning, open questions and review findings live in `requirements-design.md` and `design-review-v1.md` — not here.

1 Function

Bluetooth audio adaptor for a car AUX input. Powered from USB-C or an internal Li-ion cell.

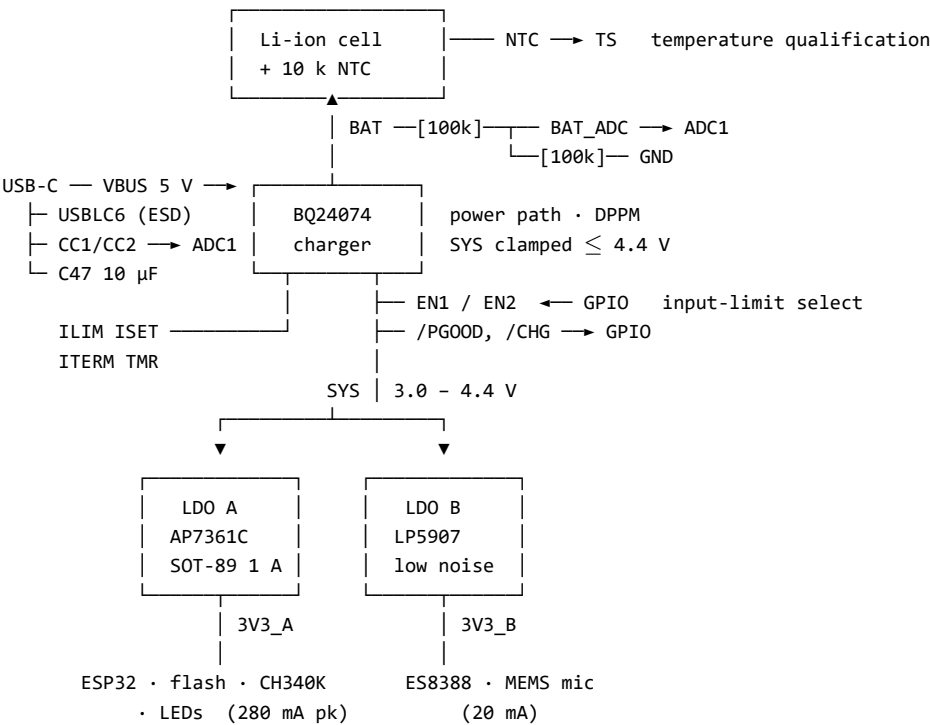
Mode	Path	Status
RX (primary)	Phone → Bluetooth A2DP → codec DAC → 3.5 mm out	Rev A hardware complete
TX	3.5 mm in → codec ADC → Bluetooth → head-phones/speaker	Rev B addition
HFP handsfree	On-board mic → codec ADC → BT; BT → DAC → jack	Rev B addition

Key constraint: must work with all phones, not just recent ones → **Bluetooth Classic (A2DP/HFP)**, not **BLE Audio**. This single requirement determines the MCU (§5).

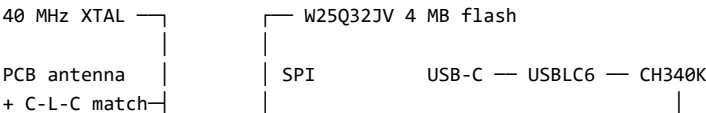
Out of scope: amplified speaker output, multipoint pairing, app control.

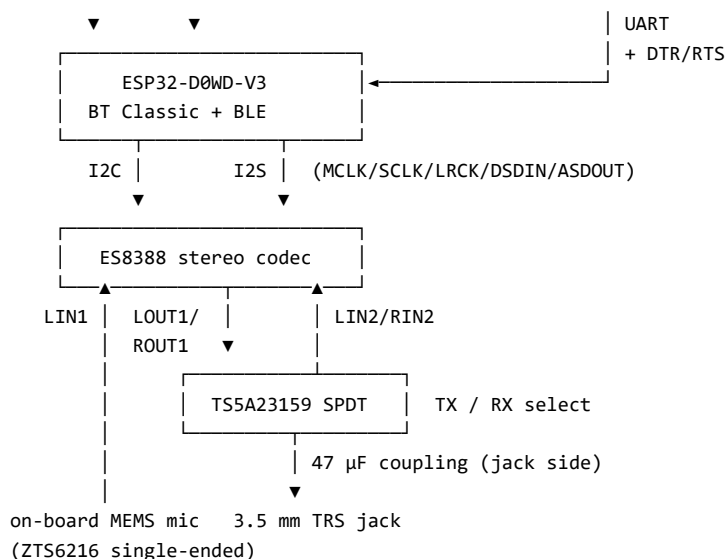
2 Architecture

Power tree



Signal chain





Rails

Rail	Source	Loads	Why separate
SYS 3.0–4.4 V	BQ24074	Both LDOs	Power path: USB when present, battery when not
3V3_A	LDO A	ESP32, flash, CH340K, LEDs	Carries RF current bursts
3V3_B	LDO B (LP5907)	ES8388, mic	Isolates a 96 dB codec from those bursts

3 Power subsystem

Linear regulation throughout. A switching converter was rejected: this is a mixed-signal audio board and the efficiency gain does not pay for the noise.

Part	Function	Key spec
BQ24074RGTR (C54313)	Power-path charger	1.5 A, SYS clamped ≤ 4.4 V, DPPM, NTC input, 10.5 V OVP
AP7361C-33Y5-13 (C460397)	LDO A — digital/RF	SOT-89-5, 1 A, ultra-low dropout
LP5907MFX-3.3 (C80670)	LDO B — codec analog	6.5 μV_{RMS} , high PSRR, no bypass cap needed
USBL6-2SC6	USB ESD	In-line on D+/D–; its VBUS pin stays at the connector, upstream of the charger
Li-ion cell + 10 k NTC	Energy	Removable. NTC bonded to cell, not PCB

VBUS reaches exactly three things: the ESD array, the charger’s IN pin, and C47. Every other load in the system hangs off SYS, including both LDOs. That is what makes the board behave identically on USB and on battery.

3.1 Power path — the five loops

In priority order. The system load always outranks the battery.

3.3 Charge programming

Resistor	Value	Formula	Result
R_ISET	1.2 k Ω 1 %	$I_{\text{CHG}} = K_{\text{ISET}}/R$, $K = 890$	742 mA (664–812 over spread)
R_ILIM	1.5 k Ω	$I_{\text{IN(max)}} = K_{\text{ILIM}}/R$, $K = 1610$	1.073 A
R_ITERM	3.0 k Ω	$I_{\text{TERM}} = 0.03 \times R_{\text{ITERM}}/R_{\text{ISET}}$	75 mA ($\approx C/27$ on a 2000 mAh cell)
R_TMR	56 k Ω	$t_{\text{MAXCHG}} = 10 \times K_{\text{TMR}} \times R$, $K = 48 \text{ s/k}\Omega$	7.5 h (5.6 h worst case)

Pre-charge falls out of the same ISET resistor: $K_{\text{PRECHG}}/R_{\text{ISET}} = 88/1200 = \mathbf{73 \text{ mA}}$, the conventional 10 % of fast charge.

742 mA is 0.37C on a 2000 mAh cell and 0.25C on a 3000 mAh one — safe for either, so this resistor does not depend on the cell decision. Use 1 % on R_ISET: TI calls this out specifically to avoid tripping the internal RISET short-circuit test.

ILIM must never be left open — an open ILIM pin disables all charging. Worth an explicit bring-up check.

R_NTC_LIN is fitted **DNP** by design: it stands in for the pack NTC if a cell without one is used, or can be replaced by a high-value linearising resistor. It must not be populated alongside a real NTC — see §3.6.

3.4 Source capability detection

The default is USB500, so detection only ever has to answer one question: **is this a $\geq 1.5 \text{ A}$ charger?** Everything else falls back safely.

CC sensing (C-to-C only). R36/R37 tap CC1/CC2 through 1 k Ω into ADC1. With $R_d = 5.1 \text{ k}\Omega$:

Source Rp	V _{CC}	Advertised
56 k Ω	0.42 V	Default USB
22 k Ω	0.94 V	1.5 A
10 k Ω	1.69 V	3.0 A

Thresholds: $>0.2 \text{ V}$ attached, $>0.66 \text{ V} \geq 1.5 \text{ A}$, $>1.23 \text{ V} = 3.0 \text{ A}$. Read **both** pins — only one carries the Rp, the other is open or is VCONN, and the pair also gives plug orientation. **No filter capacitor on the CC nodes:** the spec caps sink CC capacitance at $\approx 200 \text{ pF}$. Multisample in software instead. Re-read periodically; a source may change its advertisement at any time.

The limitation that shapes the architecture: a USB-A-to-C cable is *required* by the spec to contain a 56 k Ω Rp, so CC always reports Default USB no matter how large the charger behind it. For the expected majority case — an A-to-C cable — CC sensing tells you nothing.

BC1.2 (not fitted). The only mechanism that unlocks a USB-A wall charger, by detecting that D+ is shorted to D− through $\leq 200 \Omega$ (a Dedicated Charging Port). Needs a detector IC — BQ24392 or MAX14636 — whose integrated D+/D− isolation switch also solves the problem that the CH340K sits on that bus. Deferred; the EN1/EN2 GPIO control above is what keeps it a firmware-plus-one-IC change rather than a respin.

Guard rail. ILIM mode has no VIN-DPM, so a wrong detection has no graceful fallback. R38/R39 (100 k / 47 k) divide VBUS to USB_VBUS_SENSE — $5.0 \text{ V} \rightarrow 1.60 \text{ V}$, $5.5 \text{ V} \rightarrow 1.76 \text{ V}$ — with C53 100 nF at the tap. Firmware reverts to USB500 if VBUS sags below $\approx 4.5 \text{ V}$. The divider only draws (34 μA) when USB is present, so it costs nothing on battery.

100 nF rather than the 1 μF used on BAT_ADC: this is a protection function with a deadline. Against the divider's 32 k Ω Thévenin impedance it gives $\tau = 3.2 \text{ ms}$, so a 100 ms poll sees a collapse $\approx 30 \tau$ after it happens. A larger cap would average over a third of the reaction window.

Do not probe adaptively — stepping up to ILIM mode and watching what happens. Without VIN-DPM the failure mode is a brownout, not a graceful foldback.

Accepted risk — CC pins have no ESD protection. The USBLC6 covers D+/D− and VBUS only. CC1 and CC2 are exposed contacts in the receptacle and are directly touchable, so IEC 61000-4-2 contact discharge is a live threat. The 1 kΩ series resistors (R36/R37) protect the ADC pins but not R4/R5 or the connector node itself. They also bound a VBUS-to-CC short — which the USB-C spec requires sinks to survive — to $(5 - 3.6)/1\text{ k} \approx 1.4\text{ mA}$ of clamp injection at the ESP32, which the pin tolerates; a faulty 20 V source would push 16 mA, which it does not.

Decision (2026-08-30): not mitigated in Rev B. The fix, if it is ever wanted, is a low-capacitance TVS array on CC1/CC2 placed at the connector ahead of R36/R37 — sub-pF against the $\approx 200\text{ pF}$ sink CC budget, and low leakage so it does not corrupt the resistive advertisement. TPD4E05U06 is the obvious candidate. Revisit if field units show USB-C detection failures.

3.5 Safety timers

R_TMR = 56 kΩ gives a 7.5 h fast-charge timer (5.6 h worst case) and a 45 min pre-charge timer. Leaving TMR floating selects the internal default of 5 h typ / 4 h min; grounding it disables the timers entirely.

The timer must outlast the slowest legitimate charge. Worst case is a 3000 mAh cell charging in USB500 mode while playing: $\approx 304\text{ mA}$ of charge current against 742 mA programmed is a 0.41 reduction, so the 5.6 h timer counts at $0.41\times$ and covers 13.7 h of wall clock against an 8.9 h charge.

On expiry the part latches a fault and **/CHG flashes at $\approx 2\text{ Hz}$** . Firmware should decode that — it is also how a TS fault surfaces, and it is otherwise invisible.

3.6 Battery temperature qualification

Mandatory in hardware, not firmware. TS sources 75 μA into the pack thermistor and compares:

V_HOT = 300 mV → R_TS ≤ 4.0 kΩ → too hot ($\approx 50\text{ °C}$ on a 10 k 103AT-2)
V_COLD = 2100 mV → R_TS ≥ 28 kΩ → too cold ($\approx 0\text{ °C}$)

The valid window is therefore **4.0 kΩ – 28 kΩ**, which a bare 10 k Type-2 NTC maps onto 0–50 °C with 3 °C hysteresis. Any added network shifts those trip points: a 10 kΩ resistor in parallel with the NTC moves the hot trip to $\approx 33\text{ °C}$, and a 1 kΩ resistor pins R_TS below 1 kΩ permanently, inhibiting all charging. **Fit the pack NTC or R_NTC_LIN, never both.**

3.7 Status and state of charge

/PGOOD and /CHG are open-drain, pulled up to **3V3_A** (not SYS) so both nets are clean 3.3 V logic a GPIO can read directly — pulling them to the 4.4 V SYS rail would exceed the ESP32’s input rating.

/PGOOD	/CHG	State
Hi-Z	Hi-Z	Running on battery
Low	Low	Charging
Low	Hi-Z	Charge complete, or suspended
Low	2 Hz flash	Fault — safety timer expired, or TS out of range

R28/R29 are 330 Ω, giving $\approx 2.7\text{ mA}$ per LED at 3.3 V — visible, and inside the 5 mA the /PGOOD and /CHG open-drain outputs are specified to sink. Both LEDs are dark on battery, since the pins go high-impedance with no input present.

State of charge is voltage-based: R34/R35 form a 100 k / 100 k divider from BAT to ground, tapped as BAT_ADC (4.2 V → 2.1 V, inside ADC1’s usable 150–2450 mV window at 11 dB attenuation). It costs 21 μA continuously, $\approx 15\text{ mAh/month}$ — negligible against the cell.

c52 100 nF at the tap is not optional, and it is not impedance matching. It does two jobs:

- **Charge reservoir.** The ESP32's ADC is successive-approximation: closing the sampling switch requires an internal sample-and-hold capacitor of a few pF to charge to the input voltage *through the external source impedance*, in a few microseconds. The divider's Thévenin impedance is $100\text{ k} \parallel 100\text{ k} = \mathbf{50\text{ k}\Omega}$, five times Espressif's $<10\text{ k}\Omega$ guideline, so without a local cap each conversion drags the node down and the converter digitises the droop. At $\approx 10^4$ times the S/H capacitance, C52 supplies that charge with no measurable sag and refills through the $50\text{ k}\Omega$ between samples.
- **Low-pass filter.** $f_c = 1/(2\pi \times 50\text{ k} \times 100\text{ nF}) = \mathbf{31.8\text{ Hz}}$, $\tau = 5\text{ ms}$. BAT_ADC is a high-impedance trace crossing the board — exactly the node that collects switching and RF noise — and cell voltage moves over minutes, so heavy filtering is free accuracy. Allow $\approx 25\text{ ms}$ after power-up before trusting a reading.

The alternative was a lower-impedance divider: $10\text{ k} / 10\text{ k}$ needs no cap but draws $210\text{ }\mu\text{A}$ from the cell permanently instead of $21\text{ }\mu\text{A}$. The capacitor is the cheaper trade.

Calibrate with `esp_adc_cal` and the eFuse Vref, and multisample; raw ADC error is $\pm 6\%$, which is $\pm 250\text{ mV}$ of cell voltage and useless.

Two limits define what this can deliver. The reading is meaningless while charging — the charger is driving V_{BAT} toward 4.2 V — so firmware uses the `/PGOOD+/CHG` state instead and snaps to 100% on termination. And the Li-ion OCV curve is flat from roughly 20% to 80% SoC, so voltage alone gives $\pm 10\text{--}15\%$ there. **This is a four-segment gauge and a dependable low-battery warning, not a trustworthy percentage.** If a real percentage is ever needed over AVRCP, add a MAX17048 to the existing I²C bus.

3.8 ESP32 GPIO map (committed 2026-08-31, netlist-verified)

Complete map. The budget is **full**: one spare (GPIO33). **S** marks an ESP32 strapping pin — its level is latched at reset, so anything on it must present the right level at boot (see the implications list below). All analog is on **ADC1** because ADC2 is dead whenever the radio is on.

GPIO	Pin	Signal	Dir	S	Boot level / notes
0	23	MCLK (I ² S)	out	S	Must be HIGH at boot. Internal PU; codec MCLK input is Hi-Z at reset. Auto-boot pulls it low for download.
1	41	UART_TX (U0TXD → CH340 RXD)	out		Boot-log UART.
2	22	CHARGE_EN (→ charger /CE)	out	S	Must be LOW/float at boot — and R32 ($1\text{ k} \rightarrow \text{GND}$) holds it low, which also = charging enabled . FW drives HIGH to pause charging; any reset re-enables (fail-safe). Keeps TP4.
3	40	UART_RX (U0RXD ← CH340 TXD)	in		
4	24	AUDIO_VCC_EN (→ LDO B EN)	out		R33 pulldown → codec rail off at boot ; FW raises to power codec.
5	34	AUDIO_SW (→ TS5A23159 IN1+IN2)	out	S	Internal PU → HIGH at boot = TX by default. Do not add an external pull. FW sets LOW = RX , HIGH = TX .
6–11	31,32,33,28,29,30	SPI flash (CLK/SD0-3/CMD)	—		Reserved for VDD_SDIO flash. Not available.
12	18	MTDI (JTAG TDI)	I/O	S	VDD_SDIO voltage select — must be LOW at boot for 3.3 V flash. Pull-up R26 is DNP — do not populate . A debugger driving TDI high across reset bricks boot.
13	20	MTCK (JTAG)	I/O		J4 DNP header.
14	17	MTMS (JTAG)	I/O		J4 DNP header.

GPIO	Pin	Signal	Dir	S	Boot level / notes
15	21	MTDO (JTAG TDO)	I/O	S	Must be HIGH at boot (enables boot log). Internal PU.
16	25	EN1 (charger limit)	out (OD)		Drive open-drain (pull-up is to 4.4 V SYS). Ext 100 k→SYS = USB500 default at boot.
17	27	EN2 (charger limit)	out		Ext 100 k→GND = default LOW at boot.
18	35	I2C_CLK	I/O		To ES8388 + codec bus.
19	38	LED_GREEN	out		
21	42	CH340_EN (→ Q2 gate)	out		R40 pulldown → default ON so the bridge is powered for flashing; FW drives HIGH to disable on battery.
22	39	LED_BLUE	out		
23	36	I2C_DATA	I/O		
25	14	LRCK (I ² S)	I/O		
26	15	DSDIN (I ² S)	out		
27	16	SCLK (I ² S)	I/O		
32	12	/CHG (charger status)	in		ADC1_CH4 used as digital in.
33	13	spare	—		ADC1_CH5. Fit a no-connect flag.
34	10	BAT_ADC	in		ADC1_CH6, input-only.
35	11	ASDOUT (I ² S data in)	in		ADC1_CH7, input-only — correct home for an input-only signal.
36	5	USB_CC1	in		ADC1_CH0, input-only.
37	6	USB_CC2	in		ADC1_CH1, input-only.
38	7	/PGOOD (charger status)	in		ADC1_CH2, input-only.
39	8	USB_VBUS_SENSE	in		ADC1_CH3, input-only.

Strapping-pin implications (the ones that bite):

- **GPIO0 / GPIO2** — the boot-mode pair. Normal boot needs GPIO0 high, GPIO2 low. GPIO0 doubles as MCLK (codec input Hi-Z at reset, so safe) and the auto-boot circuit pulls it low to flash. GPIO2 = CHARGE_EN, held low by R32 — which satisfies both "low at boot" *and* "charging enabled by default." FW must only drive GPIO2 high (pause charge) *after* boot.
- **GPIO12 (MTDI)** — sets flash voltage at reset. It **must be low** (→ 3.3 V); R26 is the 1.8 V pull-up and is **DNP by design**. HW: never populate R26. FW/bench: an attached JTAG debugger can drive TDI high across reset and select 1.8 V, bricking boot — either detach during power-up or burn the XPD_SDIO_* eFuses to force 3.3 V (irreversible).
- **GPIO15 (MTDO)** must be high at reset (internal PU handles it); leave it undriven at boot.
- **GPIO5 (AUDIO_SW)** samples high at reset (SDIO-timing strap, irrelevant here). Consequence: the audio switch defaults to **TX** at boot. Harmless (no audio yet); FW drives it **low** = **RX** on init. The only HW rule: **do not tie an external pull to GPIO5**, or it fights the strap.
- General FW rule: every strapping pin used as an output must be left as an input (Hi-Z) through reset and only driven once the app starts — external pulls (or the internal strap pull) set the boot state.

Safety. Charging is inhibited outside the cell's temperature window in hardware via the NTC on TS — not in firmware. Cell is removable; device is not stored in the vehicle.

Why not LFP: 3.2 V nominal sits below the 3.3 V rail for most of its discharge, forcing a boost converter in front of the codec.

4 Audio subsystem

ES8388 stereo codec — 24-bit, 96 dB DAC dynamic range, −83 dB THD+N. Chosen over the ESP32's internal DAC for quality, and over a DAC-only part because HFP needs an ADC.

Signal	Connection
Control	I ² C, address 0x20 (CE pulled to DGND)
Audio	I ² S, codec as slave ; full duplex for HFP
Clocks	MCLK, SCLK, LRCK from ESP32
Output (RX)	LOUT1/ROUT1 → analog switch → 47 μ F → 10 Ω → jack
Input (TX)	Jack → 10 Ω → 47 μ F → analog switch → LIN2/RIN2
Input (mic)	ZTS6216 → 100 nF C0G (C41) → LIN1 (+). Pseudo-diff ready: RIN1 (−) → 100 nF C0G (C42) → GND, enabled by fitting R41 (DNP) + LINSEL=11

All four codec rails run at 3.3 V. DVDD/PVDD at 1.8 V would put the codec’s I/O in a different logic domain from the 3.3 V ESP32 — see §7.

Part	Function	Key spec
ES8388	Stereo codec	QFN-28, 0.4 mm; AVDD/HPVDD/DVDD/PVDD all 3V3
ZTS6216 (ZillTek)	MEMS microphone	LGA top-port, self-biased single-ended, 65 dB PSRR + EMI filter, SNR 65 dB(A), 1.5–3.6 V
TS5A23159DGS (C42751)	TX/RX path switch	Dual SPDT, 1 Ω R _{on} , break-before-make
PJ-3537S (C2689709)	3.5 mm TRS jack	3-pole (T/R/S) + detect switch
EEFT1E470AR (C336270)	Coupling caps C1/C21	Panasonic 47 μ F 25 V, D5×5.8 mm electrolytic
GRM31C5C1H104JA01L (C97946)	Mic coupling C41/C42	Murata 100 nF C0G 50 V 1206 — Class-1, non-microphonic (ZTS6216 requires it)
BAT54S (C408389)	TX-input clamps D7/D8	Dual series Schottky ×2; midpoint→LIN2/RIN2, ends→GND/AVDD

Output coupling. 47 μ F electrolytic (C1/C21) blocks the VMID bias. **One pair sits on the jack side of the switch and serves both directions** (RX out and TX in) — the switch stays on the codec side where both throws are at VMID, inside its 0–VCC range. Into a ≈ 10 k Ω AUX load that is a ≈ 0.34 Hz corner. The + **terminal faces the codec/VMID side**; R22/R23 (100 k Ω) bleed the jack side to 0 V to suppress insertion pops. The large value lengthens VMID settling ($\tau \approx 2$ s), so firmware mutes across power-up and TX/RX switching. **The mic coupling caps C41/C42 are C0G** (Murata C97946) — the ZTS6216 forbids Class-2 dielectrics near the mic; sizing in §8.

TX-input clamps. D7/D8 (BAT54S dual series Schottky, one per channel) clamp LIN2/RIN2 to AVDD and GND, protecting the codec inputs and the electrolytic couplers from a source driven past the rails. Off at the VMID idle point (no DC loading); they arm only beyond ≈ -0.3 V and AVDD+0.3 V, above full-scale, so normal audio is untouched. **TX input is line-level-only** (≤ 1 V_{RMS}, PGA at 0 dB): the clamps protect the codec but do not prevent clipping, and no attenuator is fitted — cutting the level would only degrade the common quieter sources.

Ground loop. Chassis ground (USB charger) and head-unit audio ground form a loop → alternator whine. R7/R8 are plain 10 Ω series resistors — **no transformer footprint is provisioned** — so **battery-only playback (USB unplugged) is the ground-loop plan of record**. Transformer isolation, if ever wanted, is a footprint change on a future spin.

5 Microcontroller & RF

ESP32-D0WD-V3. QFN-48, 0.35 mm pitch, dual-core Xtensa LX6.

Selected for firmware capability, not silicon features: it is the only Espressif part whose radio hardware supports **Bluetooth Classic**, and the only one whose SDK provides an **A2DP sink** and **HFP**. ESP32-S3/C3/C6/H2 and the entire Nordic nRF range are BLE-only. This choice follows directly from §1’s compatibility requirement.

Part	Function	Key spec
ESP32-D0WD-V3	MCU + radio	BT Classic + BLE + Wi-Fi; needs external flash
W25Q32JV	Program store	4 MB, 3.3 V (JV not JW), on VDD_SDIO
40 MHz crystal	Reference	±10 ppm required; load caps per §7
PCB inverted-F antenna + C-L-C match	2.4 GHz	50 Ω controlled impedance, 4-layer

RF front end is a deliberate learning exercise. A pre-certified module would remove the matching network, crystal trim and impedance routing — that is precisely the work being kept. Requires a NanoVNA (antenna S11) and a frequency-offset measurement to close.

Stackup: 4-layer. Signal / GND / PWR / Signal, with a solid ground plane under the RF section.

6 Programming & debug

No USB peripheral on this ESP32 variant, and no SWD (Xtensa, not ARM).

Path	Parts	Use
USB auto-program	CH340K + UMH3N dual transistor on DTR/RTS	Day-to-day flashing
Off-board serial	J3 header (DNP)	Fallback
JTAG	J4 2×5 header (DNP), ESP-PROG	Debugging

Headers are DNP — fit before a debug session. 13 test points cover I²C, mic, audio out, boot, GPIO and rails.

7 Key calculations

Power budget

Load	Rail	Current
ESP32, BT Classic streaming (avg)	A	130 mA
ESP32 transmit peak	A	250 mA
SPI flash	A	5 mA
CH340K (USB attached only)	A	10 mA
2× LED @ 330 Ω	A	8 mA
Rail A		155 mA avg · 280 mA peak
ES8388 @ 3.3 V, playback + record	B	18 mA
MEMS mic	B	0.4 mA
Rail B		≈20 mA

LDO A thermal. Bluetooth bursts last 1–2 ms; package time constant is seconds, so the junction integrates to the average, not the peak.

$$P_{\text{avg}} = (4.4 - 3.3) \times 0.155 = 0.17 \text{ W}$$

$$\text{SOT-89, } \theta_{\text{JA}} \approx 100 \text{ }^{\circ}\text{C/W} \rightarrow \Delta T = 17 \text{ }^{\circ}\text{C} \rightarrow T_{\text{j}} \approx 87 \text{ }^{\circ}\text{C at } 70 \text{ }^{\circ}\text{C ambient} \quad \checkmark$$

$$\text{SOT-23-5, } \theta_{\text{JA}} \approx 200 \text{ }^{\circ}\text{C/W} \rightarrow \Delta T = 34 \text{ }^{\circ}\text{C} \rightarrow T_{\text{j}} \approx 104 \text{ }^{\circ}\text{C} \quad \checkmark \text{ but tight}$$

LDO A dropout → usable battery. Dropout, not current, is the binding spec.

$$\text{Cutoff} = 3.3 \text{ V} + V_{\text{dropout}}(280 \text{ mA})$$

$$150 \text{ mV} \rightarrow 3.45 \text{ V} \rightarrow \sim 87 \% \text{ of cell}$$

250 mV $\rightarrow$ 3.55 V $\rightarrow$ $\sim$ 83 %
 400 mV $\rightarrow$ 3.70 V $\rightarrow$ $\sim$ 75 %
 1.2 V $\rightarrow$ 4.50 V $\rightarrow$ will not run on battery (NCP1117, Rev A – removed)

Runtime (LDO path, $\sim$ 83 % usable): 1000 mAh $\approx$ 4.8 h · 2000 mAh $\approx$ 9.7 h · 3000 mAh $\approx$ 14.5 h.

Charge current and time. R_ISET programs 742 mA, but the input limit is what actually binds in every mode except ILIM. CC delivers $\approx$ 85 % of capacity; the CV tail to a 75 mA termination adds $\approx$ 0.5 h.

Mode	Input	System	Charge	2000 mAh	3000 mAh
USB500, idle	475 mA	$\approx$ 20 mA	455 mA	4.2 h	6.1 h
USB500, playing	475 mA	171 mA	304 mA	6.1 h	8.9 h
ILIM, either	1.073 A	20–171 mA	742 mA (ISET binds)	2.8 h	3.9 h

Charger thermal. Linear, so the charger burns $(V_{IN} - V_{BAT}) \times I_{CHG}$ plus the pass-FET loss. Worst case is the start of fast charge, $V_{BAT} \approx$ 3.0 V, in ILIM mode:

$P = (5.0 - 4.4) \times 0.913 + (4.4 - 3.0) \times 0.742 = 0.55 + 1.04 = 1.59$ W
 VQFN-16 RGT, $\theta_{JA} = 44.5$ °C/W $\rightarrow \Delta T = 71$ °C $\rightarrow T_j \approx 96$ °C at 25 °C ambient ✓
 $T_j \approx 116$ °C at 45 °C ambient ✓ (reg. point 125 °C)
 USB500 mode: $P = 0.71$ W $\rightarrow \Delta T = 32$ °C $\rightarrow T_j \approx 57$ °C ✓

The input current limit is what guarantees this — raising R_ISET cannot overheat the part, because ILIM caps the current before it gets there.

Crystal load capacitance.

$CL = (C1 \times C2)/(C1 + C2) + C_{stray}$ $C_{stray} \approx$ 2–5 pF

Rev A: 18 pF $\parallel$ 18 pF $\rightarrow 9 + 3 \approx 12$ pF against a 10 pF crystal $\rightarrow$ over-loaded, frequency low
 Rev B: $\approx$ 15 pF $\parallel$ 15 pF $\rightarrow 7.5 + 3 \approx 10.5$ pF, then trim on measured offset (± 10 ppm required)

Logic-level compatibility (why all codec rails are 3.3 V).

ESP32 $V_{IH(min)} = 0.75 \times 3.3 = 2.475$ V $V_{IL(max)} = 0.825$ V
 Codec at PVDD 1.8 V: $V_{OH} \approx 1.8$ V $\rightarrow$ lands between V_{IL} and V_{IH} = undefined
 Codec abs-max input = DVDD + 0.3 V = 2.1 V $\rightarrow$ 3.3 V drive exceeds it

Audio coupling-capacitor sizing. Every audio path is AC-coupled around the codec’s VMID ($\approx$ 1.65 V) bias; each blocking cap forms a high-pass with the impedance it works into, $f_{-3dB} = 1/(2\pi R C)$. Two families:

Jack DC-blocking caps — C1/C21, 47 μ F. One shared pair on the jack side of the TX/RX switch carries **both** directions, so it is sized for the *lowest* impedance it ever sees:

RX (codec $\rightarrow$ jack): load = car head-unit AUX/line input $\sim$ 10–20 k Ω (10 k Ω = worst case)
 $f = 1/(2\pi \cdot 10 \text{ k}\Omega \cdot 47 \mu\text{F}) = 0.34$ Hz
 TX (source $\rightarrow$ codec): load = ES8388 line input 20 k Ω (external source Z is low)
 $f = 1/(2\pi \cdot 20 \text{ k}\Omega \cdot 47 \mu\text{F}) = 0.17$ Hz
 bleed R22/R23 = 100 k Ω $\parallel$ load $\rightarrow$ negligible shift

The 10 k Ω AUX case binds; 47 μ F puts even that at 0.34 Hz — no audible bass rolloff, with headroom for a head unit nearer 5 k Ω . (1 μ F would be $\approx$ 16 Hz into 10 k Ω — audibly thin; hence the electrolytic.) + terminal to the codec/VMID side; slow settling $\rightarrow$ mute across mode switches.

Mic DC-blocking caps — C41/C42, 100 nF C0G. These work into the **ES8388 ADC/PGA input impedance of 20 k Ω** (datasheet §8.3, Analog Input):

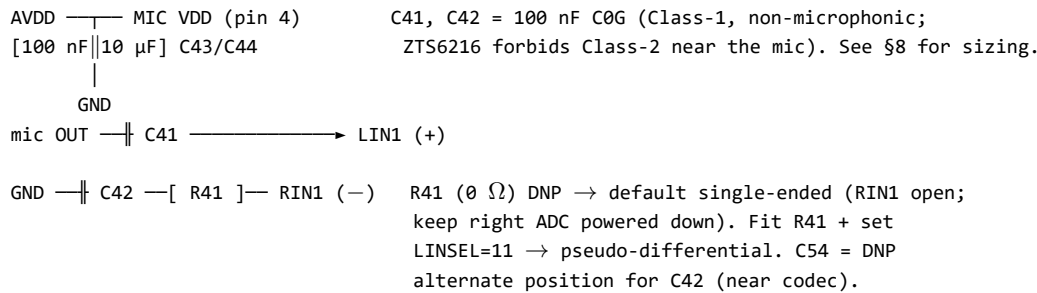
$f = 1/(2\pi \cdot 20 \text{ k}\Omega \cdot 100 \text{ nF}) = 79.6$ Hz $\approx$ 80 Hz

80 Hz is well below the HFP voice band (300 Hz–3.4 kHz) — only $\approx$ –0.3 dB at 300 Hz — so speech is unaffected, with margin if the PGA input impedance falls below 20 k Ω at gain. Both legs (C41 on +, C42 on –) use the *same* 100 nF part so the pseudo-differential high-pass stays balanced. Dielectric

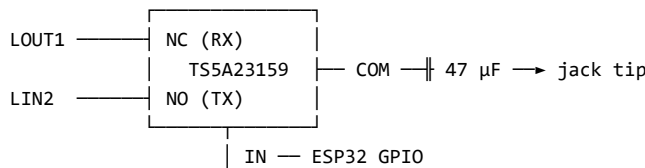
is **C0G** (Murata C97946): Class-1 ceramic is non-microphonic and voltage-linear, as the ZTS6216 requires (“no Class-2 dielectrics near the microphone”).

8 Conceptual circuits

Microphone — single-ended default, pseudo-differential-ready (ZTS6216). The ZTS6216 has one internally self-biased output, 65 dB PSRR and an on-die RF/EMI filter, so even single-ended it is defensible next to the radio (the old differential rationale was for a -45 dB-PSRR part). The board also provisions a **pseudo-differential** option: the mic drives LIN1 (+) while a matched C0G cap references RIN1 (–) to ground, so the ES8388’s differential ADC (`LINSEL=11`; default pair LINPUT1-RINPUT1) rejects common-mode supply/ground noise. SNR 65 dB(A) — adequate for HFP voice.

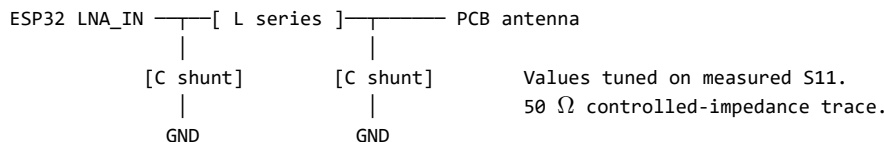


TX/RX switch — placement is critical.



Switch sits on the **codec side** of the coupling capacitor. Both throws are at VMID (≈ 1.65 V), inside the switch’s 0–3.3 V supply rails. On the jack side the signal is centred at 0 V and every negative half-cycle would clip.

RF match — C-L-C Pi, per Espressif.



9 Firmware dependencies

Hardware choices that constrain firmware, or vice versa.

Item	Note
Bluetooth Classic	Only the original ESP32 has Classic radio hardware. A2DP sink + HFP come from ESP-IDF/ESP-ADF. Drives the entire MCU selection.
Pairing persistence	Bluedroid writes bonding keys to NVS in flash automatically. Survives power-off with no battery. Requires an initialised NVS partition.
MCLK pin	ESP32 can emit MCLK only on GPIO0/1/3. GPIO1/3 are the UART, so GPIO0 — which is also the BOOT strapping pin. Safe: the codec’s MCLK input is high-Z at reset.
ADC	Use ADC1 only (battery sense). ADC2 is unusable while the radio is active. GPIO32–39 free.
Flash mode	Either works: IO2/IO3 uncrossed on the schematic 2026-08-31 (delta #4 applied), so QIO is now safe. DIO remains a fine default.
JTAG + GPIO12	MTDI is the VDD_SDIO strapping pin. A debugger driving TDI high across reset selects 1.8 V and the 3.3 V flash fails. Fix: burn <code>xpd_sdio_*</code> eFuses to force 3.3 V. Irreversible.
Codec I²C address	0x20 (CE low). ESP-ADF’s ES8388 driver default.

Item	Note
I²S full duplex TX/RX switch	Required for HFP. Configure both TX and RX channels. AUDIO_SW (GPIO5) drives both TS5A23159 control inputs (IN1+IN2 tied — the two switches are the L/R channels of one select). LOW = RX (jack ← LOUT1/ROUT1), HIGH = TX (jack → LIN2/RIN2). GPIO5 is a strapping pin: boot default is HIGH (=TX); FW must set LOW for playback on init.
Charge enable	CHARGE_EN (GPIO2) → charger /CE, active-low. R32 holds it low = charging enabled at boot; FW drives HIGH to pause charging. GPIO2 is a strapping pin (must be low at boot) — the pulldown satisfies both. A reset re-enables charging (fail-safe); the NTC still guards temperature regardless.
Deep sleep	If used for battery standby, populate the CAP2 RC network.
Charger limit	Boot default is USB500 (§3.2). Firmware may raise to ILIM mode only after confirming a ≥1.5 A source. Drive EN1 open-drain; leave both pins alone when /PGOOD is high-Z.
sequencing	
CC read	USB_CC1/USB_CC2 on ADC1 . Read both, take the one in a valid band. An A-to-C cable always reports Default USB — do not treat that as a fault.
/CHG fault	A ≈2 Hz flash is a latched fault (timer expiry or TS out of range), not "charging". Decode it; clear by toggling CE or the input.
SoC blindness	BAT_ADC is meaningless while charging. Report charger state instead, and snap to 100 % when /CHG releases.
Codec rail	AUDIO_VCC_EN is pulled low, so LDO B is off at reset . Firmware must enable it before touching the codec on I ² C/I ² S, or the ESP32 will back-power the ES8388 through its ESD diodes.
CH340K gate	Bridge is on a GPIO-controlled switch. Keep it off when USB is absent (≈10 mA of battery budget) and during any BC1.2 detection.

10 Deltas from Rev A (as built)

#	Change	Reason
1	Codec rails 1.8 V → 3.3 V; delete 1.8 V LDO, add dedicated codec LDO	Logic-level incompatibility + supply noise
2	CE pulled to DGND	Was floating — indeterminate I ² C address
3	Add 10 nF on ESP32 CAP1	Required by datasheet; was absent
4	Swap flash IO2/IO3	Crossed; corrupts QIO reads
5	Delete mic bias network (R21/R22/R23/C43/C44); add RC on mic VDD	Electret circuit on a self-biased part
6	Mic → ZTS6216 (self-biased single-ended, top-port)	ICS-40720 discontinued; ZTS6216's 65 dB PSRR + on-die EMI filter makes a single-ended connection viable next to the radio ≈0.3 mA was invisible
7	LED resistors 1 kΩ → 330 Ω	4th pole unused
8	TRRS → 3-pole TRS jack (PJ-3537S, T/R/S + detect switch)	
9	Add TX input path + analog switch	TX mode now in scope
10	Add battery charger, NTC, cell connector, VBAT sense	Battery in scope
11	NCP1117 → low-dropout LDO in SOT-89	1.2 V dropout cannot run from a cell
12	Crystal load caps 18 pF → ≈15 pF, then trim	Over-loaded; ±10 ppm required
13	Charger input limit made firmware-selectable (EN1/EN2 on GPIOs, 100 kΩ pulls)	Fixed strap cannot adapt to source capability; default USB500 boots on a flat cell
14	/PGOOD, /CHG pull-ups moved SYS → 3V3_A and routed to GPIOs	4.4 V exceeds the ESP32 input rating; status was LED-only
15	Add BAT_ADC divider, CC1/CC2 sense taps	State of charge (REQ-PWR-10) and source detection
16	Gate the CH340K on a GPIO	≈10 mA of battery budget with USB absent
17	Output coupling 1 μF MLCC → 47 μF electrolytic (C1/C21, Panasonic C336270) on the jack side; add 100 kΩ jack-side bleed (R22/R23)	Deep bass margin, one shared bidirectional cap, pop suppression
18	Add TX-input clamps D7/D8 (2× BAT54S, C408389) on LIN2/RIN2 to AVDD/GND	Protect codec inputs / electrolytic couplers from an over-range TX source
19	Mic coupling → 100 nF C0G (C41/C42, Murata C97946, 1206); provision pseudo-diff (R41 0 Ω + C54 alt, both DNP)	ZTS6216 forbids Class-2 near the mic; option to reject common-mode noise later

#	Change	Reason
20	ES8388 OUT2 → TP6 test-output couplers 1 μ F MLCC → 47 μ F electrolytic (C24/C26, Panasonic C336270), matching C1/C21; + terminal to the codec/VMID side	Removes Class-2 ceramic coupling distortion; one shared coupler part across the board

Still open: cell choice (2000 mAh pouch vs 3000 mAh 18650) and therefore charge time; whether to fit BC1.2 detection; ground-loop mitigation (transformers vs battery-only playback); crystal offset measurement; RF match tuning; BOM line consolidation.

Accepted risks: CC pins carry no ESD protection (§3.4) — decided 2026-08-30, not mitigated in Rev B.

Micro-sheet status (§3 describes the target). Tracking the 2026-08-31 micro-sheet review ([design-review-micro.m](#)) and the fixes applied since.

Applied on the schematic (verified against netlist 2026-08-31):

- ✓ **UART TX/RX crossover fixed** — UART_RX = {ESP32 U0RXD, CH340 TXD}, UART_TX = {U0TXD, CH340 RXD}. Board can now be flashed. (was S1)
- ✓ **Flash IO2/IO3 uncrossed** — SPIWP(GPIO9)→IO2, SPIHD(GPIO10)→IO3. QIO-safe now. (delta #4, was S1)
- ✓ **Digital rail unified** — micro +3V3 renamed to +3V3_SYS, tied to the LDO A output (U7.5). The stray +3V3 symbol on the ESP32 EN pull-up (R11) was also switched to +3V3_SYS, so EN is no longer floating.
- ✓ **CH340K load switch drawn** (delta #16) — Q2 AO3401A (SOT-23, LCSC C15127) high-side P-FET: source +3V3_SYS → drain CH340_VCC → CH340 VCC/V3 + C19. Gate CH340_EN on **GPIO21**, with R40 100 k Ω **pull-down = default-ON** (so the ROM bootloader is powered for flashing with no firmware). Firmware drives GPIO21 **high to disable** the bridge on battery. Note: when off, the bridge's UART/DTR/RTS pins can still back-feed its dead VCC through clamp diodes — tristate the ESP32 UART pins when disabling, or accept the small leakage.
- ✓ **Full ESP32 interface wired** (§3.8) — EN1/EN2//PGOOD//CHG/BAT_ADC/USB_CC1/USB_CC2/USB_VBUS_SENSE/AUDIO_VCC_EN/CHARGE_EN all land on GPIOs; all four analog senses on ADC1. The TX/RX switch (AUDIO_SW, GPIO5) drives both TS5A23159 control inputs (delta #9 now controllable). Charge enable (CHARGE_EN, GPIO2) is now firmware-controllable.

Still to do on the micro sheet:

- **Crystal load caps still 18 pF** (delta #12 not applied; target $\approx$ 15 pF then trim).
- **Firmware LEDs D1/D3 still 1 k Ω** (delta #7 hit the power LEDs only; blue $\approx$ 0.3 mA, invisible).
- **GPIO33 spare** — fit a no-connect flag so it doesn't ERC.
- **Schematic hygiene:** stale library-symbol caches, off-grid endpoints, annotation errors — ERC cleanup due.

Rev A→B deltas still unapplied: #7 (LED resistors — micro sheet only), #12 (crystal caps). Deltas #4, #9, #13–17 are now on the schematic.

Audio-sheet status (§6 describes the target). Tracking the 2026-09-07 audio review ([design-review-audio.md](#)), re-audited from a fresh netlist 2026-09-08.

Applied on the schematic (verified against netlist 2026-09-08):

- ✓ **I²C pull-ups moved to +3V3_AUDIO**, dropped to 1 k Ω (R19/R20) — they die with the codec rail, so gating AUDIO_VCC_EN no longer back-powers the ES8388. (was S2)

- ✓ **L/R un-swapped** — COM1→ROUT1/RIN2 (Right)→Ring, COM2→LOUT1/LIN2 (Left)→Tip, matching J1's TIP_LEFT/RING_RIGHT. Standard stereo orientation in both RX and TX. (was S2)
- ✓ **Jack** → **PJ-3537S 3-pole TRS** with detect switch (delta #8). (was S3)
- ✓ **47 μ F electrolytic couplers** (C1/C21) on the jack side of the switch, correct polarity (+ to VMID side), with 100 k Ω jack-side bleed (R22/R23) (delta #17).
- ✓ **TX-input clamps** D7/D8 (2 $\times$ BAT54S, C408389) — series-clamp geometry verified: common→signal, anode→GND, cathode→AVDD; off at VMID, no DC loading (delta #18).
- ✓ **Mic coupling** → **100 nF C0G** (C41/C42, Murata C97946, 1206), sized in §8; pseudo-diff option provisioned with R41 (0 Ω) and C54 both DNP (delta #19, #6).

Open / notes on the audio sheet:

- ✓ **TX input level** — **decided line-level-only** (2026-09-08). Expect $\leq 1 V_{\text{RMS}}$ with the PGA at 0 dB; ES8388 full-scale in = AVDD/3.3 $\approx 1.0 V_{\text{RMS}}$. No attenuator — it would halve the common quieter sources (0.316 V_{RMS} line-out $\rightarrow$ 0.16 V) and cost SNR; the BAT54S clamps are damage protection only, so a hotter source clips softly, not a fault. (A7 closed)
- **RIN1 floats in the default build** — with R41 DNP, the mic – reference is unconnected. Benign as long as firmware keeps the right ADC powered down; fitting R41 (which also enables pseudo-diff) removes it. Deliberate provisioning choice, not a defect.
- **Firmware mute** across power-up and TX/RX switching — the 47 μ F couplers give slow ($\sim 1\text{--}5$ s) VMID settling. (A8)
- **Remove D7/D8 TX-input clamps** (decided 2026-09-16) — LIN2/RIN2 are internal line-level nets with no external connector, so the codec's own ESD structures suffice; the BAT54S junction capacitance (~ 10 pF) and reverse leakage degrade the high-impedance analog input for no protective benefit. **Schematic edit pending** (done by hand); delta #18 to be struck and §6/§9 clamp text removed once applied.

11 Manufacturing

JLCPCB, 4-layer, **Economic PCBA** with the ESP32 excluded from assembly and hand-populated (it is the only Standard-Only part). All other parts verified Economic-tier.

Constraints: single-sided placement (satisfied — nothing on the back is assembled), 0402 minimum passive, 0.4 mm minimum IC pitch.

Build: 5 assembled + 5 bare. Target \$20 USD landed per unit.

Hand assembly: 0.35 mm-pitch QFN requires a laser-cut stencil, plugged/capped via-in-pad on the thermal pad (open vias wick solder and degrade the RF ground), and $\approx 50\text{--}70\%$ paste coverage on the exposed pad.

12 References

Local (hw/datasheets/, hw/app notes/)

- ESP32 datasheet · ESP32 Hardware Design Guidelines · ESP32 Technical Reference Manual
- ES8388 datasheet rev 5.0 · ES8388 User Guide
- W25Q32JV datasheet rev G
- WCH CH340 datasheet v3B ([WCH-CH340-datasheet-v3B.pdf](#)) — TXD=output, RXD=input; V3→VCC at 3.3 V

- ESP32-LyraT v4.3 schematic — reference design for the codec section
- AN043 2.4 GHz PCB Antenna · Antennas for IoT

External

- JLCPCB PCB assembly capabilities
- ESP-ADF ES8388 driver
- esptool — set flash voltage eFuses

Project

- requirements-design.md — requirements, constraints, traceability
- design-review-v1.md — netlist audit and findings (initial)
- design-review-micro.md — micro-sheet review, 2026-08-31
- battery-power-proposal1.md — battery architecture
- mems-microphone-primer.md — mic theory and selection

13 Revision history

Rev	Date	Change
A	2025-04-05	As-built schematic, commit 369c5f9
B	2026-08-24	This document created. Scope decisions resolved; 12 deltas in §10
B	2026-08-30	§3 expanded to cover the full charger configuration: power-path loops, EN1/EN2 strapping, charge programming, source detection, safety timers, TS qualification, SoC and GPIO map. Deltas 13–16
B	2026-08-30	Power sheet complete: <code>R_ITEM/R_TMR</code> populated, LEDs to 330 Ω , ADC anti-droop caps and VBUS sense divider added. CC ESD recorded as an accepted risk
B	2026-08-31	Micro-sheet review (<code>design-review-micro.md</code>): found UART TX/RX swap (S1) and confirmed flash IO2/IO3 still crossed; corrected the §9 “flash fixed” claim and the §3.8 GPIO map (added TX/RX switch, fixed the free-output count); expanded §10 with the undrawn micro↔power interface and unapplied deltas
B	2026-08-31	Micro-sheet fixes landed: UART crossover, flash IO2/IO3, rail unified to <code>+3V3_SYS</code> (EN pull-up reconnected), and CH340K high-side load switch drawn (Q2 AO3401A + R40 100 k Ω pull-down, gate <code>CH340_EN</code> on GPIO21, default-ON). Netlist-verified.
B	2026-08-31	Full ESP32 GPIO map committed (§3.8) with strapping-pin FW/HW implications: interface signals wired to the ESP32 (EN1/EN2, /PGOOD, /CHG, ADC senses, <code>AUDIO_VCC_EN</code> , <code>CHARGE_EN</code> →GPIO2), <code>AUDIO_SW</code> →GPIO5 driving both TS5A23159 inputs (LOW=RX, HIGH=TX). Verified TS5A23159 pinout against TI datasheet — COM=jack/NC=codec-out/NO=codec-in, both channels symmetric (a prior review note claiming asymmetry was wrong, now withdrawn).
B	2026-09-03	Design overview converted from Markdown to LaTeX (Tectonic/XeTeX build) for easier manual editing. Content unchanged.
B	2026-09-07	Audio sheet re-reviewed from a fresh netlist (<code>design-review-audio.md</code>). Folded confirmed audio decisions into §2/§6/§8/§9/§10: single-ended ZTS6216 mic (was differential SPH8878), 47 μ F electrolytic jack-side couplers C1/C21 + R22/R23 bleed (was 1 μ F MLCC), PJ-3537S TRS jack, L/R and I ² C-pull-up-rail fixes. Corrected the stale “isolation-transformer footprints provisioned” claim — R7/R8 are plain 10 Ω , battery-only is the ground-loop plan. Also recorded the earlier withdrawal of the ES8388 OUT1-vs-OUT2 line/headphone distinction (no such split in the datasheet).
B	2026-09-08	Audio caps + clamps finalized against the netlist: TX-input clamps to 2× BAT54S (D7/D8, common→signal / anode→GND / cathode→AVDD); mic DC-blocking caps to 100 nF C0G (Murata C97946, 1206) with a provisioned pseudo-differential option (R41 0 Ω + C54 alt, both DNP). Added §8 coupling-cap sizing calculations — jack couplers vs the car-AUX load (0.34 Hz into 10 k Ω) and mic caps vs the ES8388 20 k Ω ADC input (80 Hz). I ² C pull-ups noted at 1 k Ω . TX input level decided line-level-only (PGA 0 dB, no attenuator; clamps = damage protection) — A7 closed.

Rev	Date	Change
B	2026-09-16	Pre-layout BOM pass on the audio sheet: ES8388 OUT2 → TP6 test-output couplers C24/C26 → 47 μ F electrolytic (Panasonic C336270, matching C1/C21, + to codec/VMID side), removing Class-2 ceramic coupling distortion (delta #20); c53 VBUS-sense filter 1206 → 0402 (house part C1525); c41/c42 mic couplers footprint aligned to plain c_1206_3216Metric (matching C54). Decided to remove the D7/D8 TX-input clamps (LIN2/RIN2 are internal line-level nets, no external exposure) — schematic edit pending. All PCB footprints now placed in initial positions; V1 routing pending.